AN AGENT-BASED MODEL OF CULTURAL CHANGE FOR A LOW-CARBON TRANSITION

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INTRODUCTION

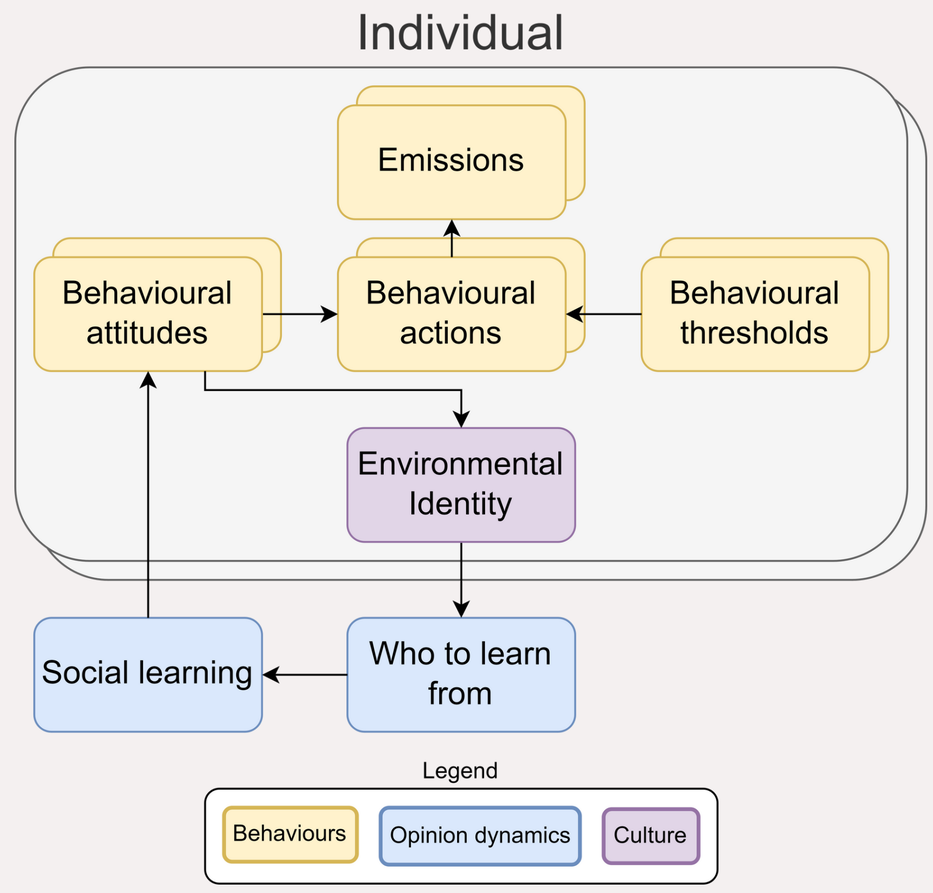
Meeting climate goals requires radical changes in the consumption behaviours of individuals. Theoretical models of behavioural change can provide insight into what barriers and drivers exist to the adoption of low-carbon lifestyles. A crucial element in the study of a transition towards low-carbon lifestyles is how quickly behaviours in a population, and even culture itself, will change. Moreover, a multi-behavioural perspective is important due to the breadth of lifestyle changes required for deep decarbonisation.

Culture may change slowly over time thereby limiting the effectiveness of climate policy e.g. information provision or carbon pricing or contributing towards lock-in of brown behavioural alternatives. We strive to generate awareness of what roadblocks to decarbonisation may appear due to interactions between culture and behavioural diffusion.

RESEARCH QUESTIONS

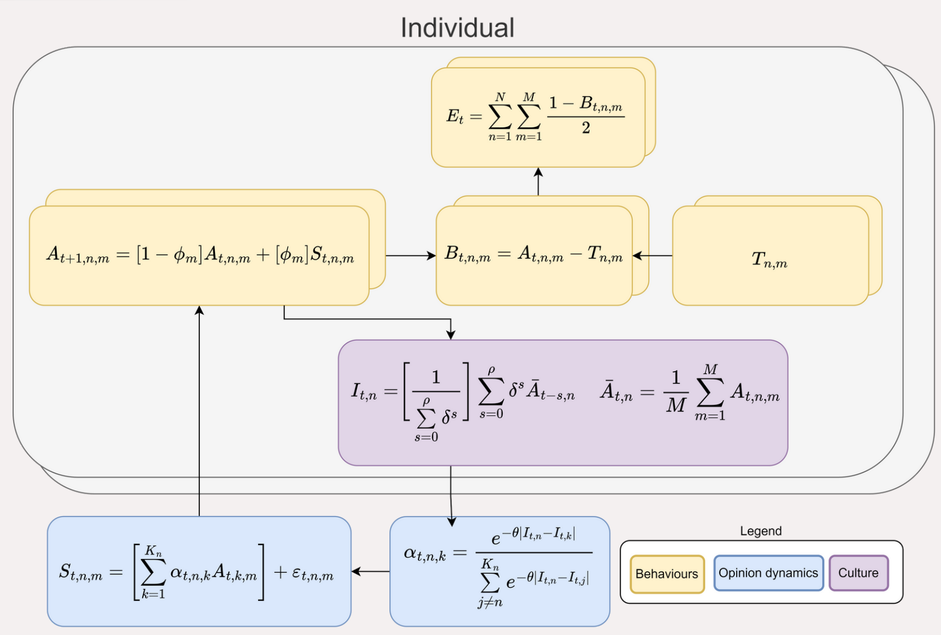
- How does pro-environmental diffusion of behaviour interact with longer-term cultural evolution when considering the immediacy of climate change?
- Does the interdependency of intra-personal behaviours spur on or inhibit diffusion?

MODEL OVERVIEW

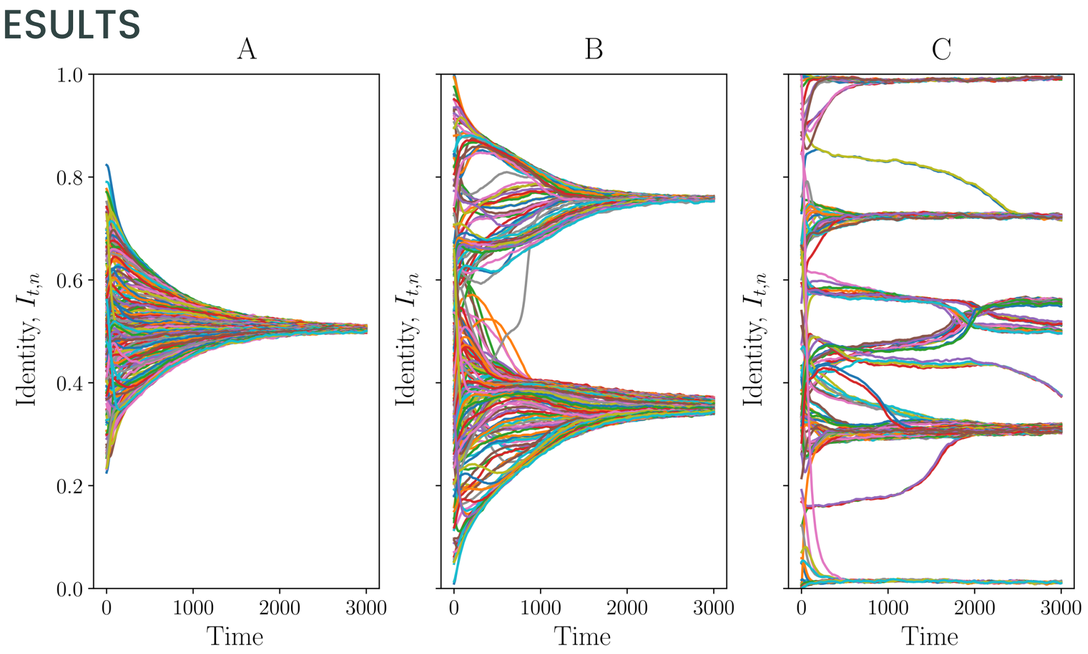


MODEL

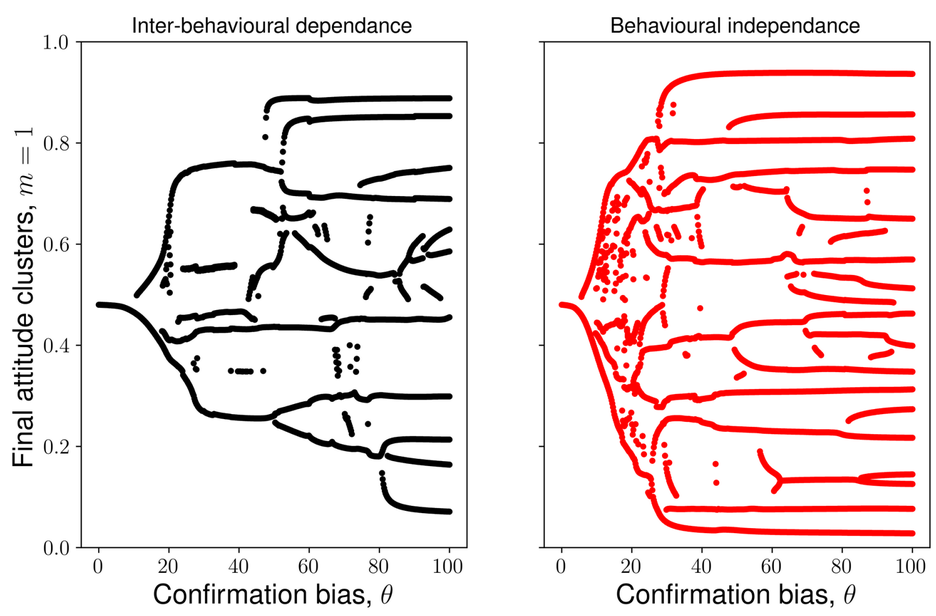
- Favourable attitudes towards a behavioural option are not sufficient, one must also have the means or control to perform them.
- Culture as socially transmitted information.
- Slower, longer-term change process of environmental identity driven by a faster behavioural diffusion process.
- Social information exchange in a Watts–Strogatz graph.
- Individuals aggregate the weighted influence of all neighbours in their ego network subject to a confirmation bias.
- Stronger environmental identity makes pro-environmental behaviour more frequent.



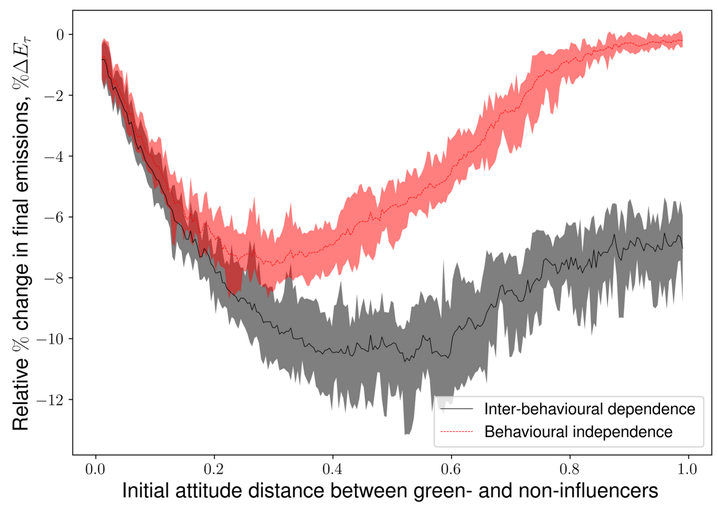
RESULTS



Increasing (A to C) initial attitude polarisation and confirmation bias creates strands of "identity bubbles" and slower consensus formation



Environmental identity stimulates the convergence of attitudes, relative to the behavioural inter-dependence case.



Stronger decarbonisation impact, and over a broader range of attitude distances, of green influencers on individuals for the presence of culture (inter-behavioral dependence)

CONCLUSIONS

- Environmental identity helps stimulate consensus formation in behavioural choices.
- The degree of behavioural decarbonisation of individuals in the social network strongly depends on the initial preferences in behavioural attitudes and thresholds.
- Green influencers fail to achieve deep behavioural decarbonisation through solely voluntary action. This highlights the need for further research into policy interventions in the form of tailored information provision, and the impact of culture on carbon pricing that would reduce barriers to performing greener behaviours.

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Paper submitted for review

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